EXACT FIBER ANALYSIS OF KINETIC-BALANCE SPECTRAL POLLUTION FOR THE RADIAL DIRAC OPERATOR

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Abstract. Galerkin discretizations of the radial Dirac operator are notorious for spectral pollution: spurious eigenvalues inside the essential gap that kinetic-balance prescriptions are designed to suppress. We show that for the discretized free operator this phenomenon admits an exact, completely elementary analysis. The free Dirac matrix built on any scalar basis decomposes into two-dimensional fibers indexed by the singular values of the coupling matrix, and on each fiber the sector leakage of the balanced, split and anti-balanced trial families is an explicit algebraic function of the dimensionless fiber parameter $t = \sigma/2c$: the balanced families leak into the wrong spectral sector as t^3, the split family as t, and the anti-balanced family as $2t$, with exact constants. The global conformity defect of an untruncated single-family prescription equals $\max_j \ell_j\sqrt{1-\ell_j^2}$ exactly, and the full dual-kinetic-balance pair space is exactly conforming. For any bounded potential we prove a pollution dichotomy: the balanced, positronic and split prescriptions produce no Galerkin eigenvalue in the potential-free window $(-c^2+v,\, c^2-v)$, for any potential of norm at most v and with no smallness condition, while the anti-balanced prescription is guaranteed to produce spurious eigenvalues there, one for each fiber whose Rayleigh value $c^2(5t^2-1)/(1+t^2)$ sits deep enough inside the gap. A companion cusp lemma explains the approximation-theoretic side: expansion coefficients of the Coulomb–Dirac cusp profile r^γ in a generalized Laguerre basis with prefactor power μ decay with the universal exponent $-(1+\gamma)$ regardless of μ, the mismatch entering only through the constant $\Gamma(\mu+\gamma+1)/\Gamma(\mu-\gamma)$, which vanishes precisely at the matched prefactor. All statements are verified numerically to machine precision in a certified Laguerre discretization, and the measured eigenvalue-convergence law $N^{-2\gamma}$, the convergence of the continuum-relative form bound, and the trajectory of the anti-balanced impostor across the gap at point Coulomb coupling are reported as the empirical boundary of the theory, together with the two statements they support as conjectures.

1. Introduction

The Dirac operator is unbounded below, and naive Rayleigh–Ritz discretization of its eigenvalue problem produces spurious eigenvalues inside the spectral gap — spectral pollution — rather than variational collapse

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alone. The phenomenon has two well-developed bodies of theory. On the abstract side, pollution for operators with gaps is characterized by the geometry of the trial spaces relative to the spectral splitting: the min-max principles of Talman [1] and Dolbeault, Esteban and Séré [2] hold up to a coupling ceiling made precise by Morozov and Müller [3], and the projector conditions of Lewin and Séré [4] (see also [5]) give general criteria for pollution-free limits. On the practical side, relativistic atomic structure has long relied on kinetic-balance constructions — restricted kinetic balance, and the dual kinetic balance of Shabaev and co-workers [6] — whose empirical success is well documented [7, 8], as is the spurious-state phenomenology of unbalanced bases [9, 10, 11].

What has been missing between the two is a setting in which the pollution mechanism can be computed exactly rather than bounded abstractly or observed numerically. This paper provides such a setting. The observation is elementary: the discretized *free* radial Dirac operator on any scalar basis block-diagonalizes into 2×2 fibers indexed by the singular values of the discrete coupling matrix, and every kinetic-balance prescription places exactly one or two trial vectors in each fiber. All the relevant quantities — sector leakages, conformity defects, and the free Rayleigh values whose position inside or outside the gap decides pollution — then reduce to explicit algebra in the single dimensionless parameter $t = \sigma/2c$. The resulting statements are sharp, carry exact constants, and are verified to machine precision in the certified discretization of the companion paper [13], where the same structure underlies a point-nucleus solver accurate to 10^{-14} with tens of basis functions.

Sections 2–4 develop the fiber decomposition, the exact per-mode leakage laws, and the exact global conformity defect. Section 5 proves the pollution dichotomy for bounded potentials. Section 6 proves the cusp lemma: the exact representation of the point-nucleus ground state in the matched Laguerre basis, and the universal-exponent rate dichotomy for mismatched prefactors. Section 7 reports the numerical evidence at the boundary of the theory — the measured $N^{-2\gamma}$ eigenvalue law, the convergence of the continuum-relative form bound where its discrete-relative counterpart provably diverges, and the trajectory of the anti-balanced impostor across the gap at point Coulomb coupling — and states the two conjectures this evidence supports.

Throughout, Hartree atomic units are used, c denotes the speed of light, and all spaces are real.

2. The discrete operator and its fiber decomposition

On the channel $\kappa \in \mathbb{Z}\setminus\{0\}$ the radial Dirac operator with potential V acts on two-component functions as

$$h_\kappa = \begin{pmatrix} V + c^2 & c\,\mathcal{O}_+ \\ c\,\mathcal{O}_- & V - c^2 \end{pmatrix}, \qquad \mathcal{O}_\pm = \mp\frac{d}{dr} + \frac{\kappa}{r}. \tag{1}$$

Let $\{f_j\}_{j=1}^N$ be an orthonormal scalar family in $L^2(0,\infty)$ contained in the common domain of $\mathcal{O}_\pm$, with vanishing boundary terms, so that $\langle f_i,\, \mathcal{O}_+ f_j\rangle = \langle \mathcal{O}_- f_i,\, f_j\rangle$. Expanding both components in this family, the discretized free operator ($V \equiv 0$) is the symmetric matrix

$$H_0 = \begin{pmatrix} c^2 I & cK \\ cK^T & -c^2 I \end{pmatrix} \in \mathbb{R}^{2N\times 2N}, \qquad K_{ij} = \langle f_i,\, \mathcal{O}_+ f_j\rangle. \tag{2}$$

Proposition 2.1 (fiber decomposition)**.** *Let $K = U\Sigma V^T$ be a singular value decomposition, $\Sigma = \mathrm{diag}(\sigma_1,\dots,\sigma_N)$, and write $e_1^{(j)} = (u_j, 0)$, $e_2^{(j)} = (0, v_j)$. Then $\mathbb{R}^{2N} = \bigoplus_j W_j$ with $W_j = \mathrm{span}\{e_1^{(j)}, e_2^{(j)}\}$, each W_j is H_0-invariant, and in the basis $(e_1^{(j)}, e_2^{(j)})$ the restriction of H_0 to W_j is the fiber matrix*

$$h(\sigma_j) = \begin{pmatrix} c^2 & c\sigma_j \\ c\sigma_j & -c^2 \end{pmatrix}. \tag{3}$$

Consequently $h(\sigma)^2 = E(\sigma)^2 I$ with $E(\sigma) = \sqrt{c^4 + c^2\sigma^2}$, the spectral projectors $\Lambda_\pm$ of H_0 decompose fiberwise, and $|H_0| = \bigoplus_j E(\sigma_j)\, I_{W_j}$ is scalar on each fiber.

Proof. From $Kv_j = \sigma_j u_j$ and $K^T u_j = \sigma_j v_j$ we obtain $H_0 e_1^{(j)} = c^2 e_1^{(j)} + c\sigma_j e_2^{(j)}$ and $H_0 e_2^{(j)} = c\sigma_j e_1^{(j)} - c^2 e_2^{(j)}$, which is invariance with restriction (3); orthogonality of the u_j and of the v_j gives the orthogonal direct sum. The square of (3) is $(c^4 + c^2\sigma_j^2)I$ by direct multiplication, and functions of a block-diagonal symmetric matrix are block-diagonal, which yields the statements about $\Lambda_\pm$ and $|H_0|$. □

The kinetic-balance prescriptions place, in the fiber coordinates (x, y) of W_j and with $t_j = \sigma_j/2c$, the trial directions

$$u_{\mathrm{bal}} = (1, t), \quad u_{\mathrm{spl}} = (1, 0), \quad u_{\mathrm{pos}} = (-t, 1), \quad u_{\mathrm{anti}} = (t, 1). \tag{4}$$

Indeed, the balanced electronic pair $(f, \frac{1}{2c}\mathcal{O}_- f)$ of restricted and dual kinetic balance has, for $f = u_j$, lower-component coefficients $K^T u_j/2c = t_j v_j$, giving $(1, t_j)$; the physical dual-balance positronic pair $(-\frac{1}{2c}\mathcal{O}_+ f, f)$ gives $(-t_j, 1)$ for $f = v_j$; the sign-flipped (anti-balanced) variant gives $(t_j, 1)$; and the naive equal-basis prescription contributes the split directions. Linearity implies that the span of each untruncated family over the whole scalar basis equals the span of its per-fiber directions.

3. Exact per-mode leakage

Theorem 3.1 (per-mode leakage laws)**.** *Fix a fiber with parameter $t > 0$, let $w = \sqrt{1 + 4t^2}$ and $\rho = 2t/(1 + w)$, and let $\ell(u)$ denote the norm of the projection of the normalized trial vector u onto the wrong spectral sector of $h(2ct)$ (negative sector for the electronic families, positive sector for the*

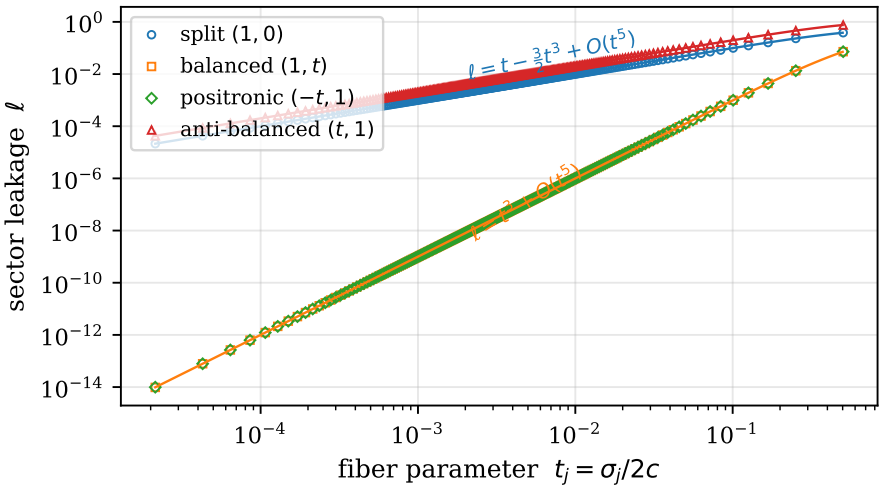


FIGURE 1. Measured per-mode sector leakages of the four trial families in a 240-function Laguerre discretization of the $\kappa = -1$ channel (symbols; one point per SVD fiber), overlaid with the parameter-free laws of Theorem 3.1 (lines). The balanced and positronic families coincide, as the theorem states; agreement is at machine precision over fourteen decades.

positronic ones). Then, exactly,

$$\ell(u_{\mathrm{spl}}) = \frac{\rho}{\sqrt{1+\rho^2}} = t - \tfrac{3}{2}t^3 + O(t^5), \tag{5}$$

$$\ell(u_{\mathrm{bal}}) = \ell(u_{\mathrm{pos}}) = \frac{4t^3}{(1+w)^2\sqrt{1+\rho^2}\sqrt{1+t^2}} = t^3 + O(t^5), \tag{6}$$

$$\ell(u_{\mathrm{anti}}) = \frac{t+\rho}{\sqrt{1+\rho^2}\sqrt{1+t^2}} = 2t - 3t^3 + O(t^5). \tag{7}$$

Proof. The normalized eigenvectors of $h(2ct)$ are $v_+ = (1,\rho)/\sqrt{1+\rho^2}$ and $v_- = (-\rho,1)/\sqrt{1+\rho^2}$ with eigenvalues $\pm E$, since $\rho = 2t/(1+w)$ satisfies $c\,(2ct) = \rho\,(E + c^2)$ with $E = c^2 w$. The leakages are the absolute inner products of the normalized trial vectors with the wrong-sector eigenvector. For the split family this gives (5) directly. For the balanced family, $|\langle v_-,\,(1,t)\rangle| = (t-\rho)/\sqrt{1+\rho^2}$, and the algebraic identities

$$t - \rho = \frac{t(w-1)}{w+1} = \frac{4t^3}{(1+w)^2}, \tag{8}$$

the second from $(w-1)(w+1) = 4t^2$, resolve the absolute value (the right side of (8) is positive) and give (6); the positronic case is the same computation with v_+ and $(-t,1)$. The anti-balanced case is $|\langle v_+,\,(t,1)\rangle| = (t+\rho)/\sqrt{1+\rho^2}$. The expansions follow from $w = 1 + 2t^2 + O(t^4)$. □

Figure 1 shows the four laws against the measured leakages in the actual Laguerre discretization: the reduction is exact in the discrete setting, so the agreement is limited only by machine precision. Theorem 3.1 quantifies the geometry behind kinetic balance: the balanced constructions are conforming to *third* order in t, the split and anti-balanced ones only to first order, with the anti-balanced constant twice the split one — and this discriminates the sign convention of the dual-balance family [6], whose flipped variant is anti-balanced.

4. The global conformity defect

Theorem 4.1 (exact global defect)**.** *Let Q have orthonormal columns spanning an untruncated single-family trial space (one direction of (4) per fiber), let $\Pi = QQ^T$, and let $\delta = \|(I-\Pi)\Lambda_+\Pi\|_2$. Then, exactly,*

$$\delta = \max_j \ \ell_j\sqrt{1-\ell_j^2}, \tag{9}$$

with ℓ_j the corresponding leakage of Theorem 3.1, and the same value is obtained with Λ_- in place of Λ_+.

Proof. Π, $\Lambda_\pm$ and hence $(I-\Pi)\Lambda_+\Pi$ are block-diagonal with respect to the fiber decomposition of Proposition 2.1, and the norm of a block-diagonal matrix is the maximum of the block norms. On a fiber with unit trial vector b and $g = \langle v_+, b\rangle$, one computes $(I - bb^T)\, v_+ v_+^T\, b = g\,(v_+ - gb)$, whose squared norm is $g^2(1-2g^2+g^2) = g^2(1-g^2)$ using $\|v_+\| = \|b\| = 1$. Since $\langle v_+, b\rangle^2 + \langle v_-, b\rangle^2 = 1$ in the fiber, $g^2 = 1-\ell^2$ and the block norm is $\ell\sqrt{1-\ell^2}$; the expression is symmetric under $\ell^2 \leftrightarrow 1-\ell^2$, which proves the Λ_- statement. □

Corollary 4.2 (exact conformity of the dual-balance pair space)**.** *The untruncated dual-kinetic-balance pair space, spanned by both $(1, t_j)$ and $(-t_j, 1)$ in every fiber, satisfies $(I-\Pi)\Lambda_\pm\Pi = 0$ exactly.*

Proof. The two directions span each two-dimensional fiber, so Π restricts to the identity on every fiber, and $\Lambda_\pm$ preserve fibers. □

Corollary 4.2 sharpens the empirical picture of dual kinetic balance: the prescription itself is *exactly* conforming to the free splitting; any nonzero defect observed in practice is entirely an effect of truncating the family, and it grows with the basis size only because the largest singular value does — in the Laguerre families used here, $\sigma_{\max} \approx 0.64\,\beta N$ (measured exponent 0.992), so the danger fibers $t = O(1)$ are always the highest modes.

5. A pollution dichotomy for bounded potentials

For a bounded potential the entire pollution question for single-family prescriptions reduces to the position of the free Rayleigh values.

Theorem 5.1 (pollution dichotomy)**.** *Let $V_d \in \mathbb{R}^{N\times N}$ be symmetric with $\|V_d\|_2 \le v$ — in particular the compression of a bounded multiplication operator with $\|V\|_\infty \le v$ onto the orthonormal family — and let $W = \mathrm{diag}(V_d, V_d)$. Write $G_v = (-c^2 + v,\, c^2 - v)$ and, per fiber,*

$$R_{\mathrm{bal}} = c^2\frac{1+3t^2}{1+t^2}, \quad R_{\mathrm{spl}} = c^2, \quad R_{\mathrm{pos}} = -R_{\mathrm{bal}}, \quad R_{\mathrm{anti}} = c^2\frac{5t^2-1}{1+t^2}. \tag{10}$$

Then:

(a) $R_{\mathrm{bal}} - c^2 = 2c^2t^2/(1+t^2) \ge 0$*; R_{anti} lies strictly inside $(-c^2, c^2)$ for all $0 < t < 1/\sqrt{2}$ and equals the exact gap midpoint 0 at $t = 1/\sqrt{5}$.*

(b) $\mathrm{spec}(H_0 + W) \cap G_v = \emptyset$.

(c) *Every Galerkin eigenvalue of $H_0 + W$ on the balanced family is $\ge c^2 - v$; on the positronic family, $\le -c^2 + v$; on the split family, within v of c^2. None of these prescriptions produces an eigenvalue in G_v, for any bounded potential and with no smallness condition.*

(d) *On the anti-balanced family the sorted Galerkin eigenvalues lie within v of the sorted $R_{\mathrm{anti}}(t_j)$; every fiber with $|R_{\mathrm{anti}}(t_j)| < c^2 - 2v$ therefore forces a Galerkin eigenvalue inside G_v, where by (b) the parent matrix has no spectrum: a genuine spurious eigenvalue, with predicted location.*

Proof. (a) is direct computation from (10): $c^2 - R_{\mathrm{anti}} = c^2(2-4t^2)/(1+t^2)$ and $R_{\mathrm{anti}} + c^2 = 6c^2t^2/(1+t^2)$. For a matrix Q with orthonormal columns, $\|Q^TWQ\| \le \|W\| = \|V_d\| \le v$ (Cauchy–Schwarz on unit vectors Qx), and for the compression of a multiplication operator $\|V_d\| \le \|V\|_\infty$ by the same argument in L^2. Q^TH_0Q is diagonal with entries (10) by Proposition 2.1 and the proof of Theorem 4.1. Weyl's eigenvalue perturbation inequality, $|\lambda_k(D+P) - \lambda_k(D)| \le \|P\|$ for symmetric matrices, applied with $D = H_0$, $P = W$ gives (b), since $\mathrm{spec}(H_0) = \{\pm E(\sigma_j)\}$ avoids $(-c^2, c^2)$; applied with $D = Q^TH_0Q$, $P = Q^TWQ$ it gives (c) and the pairing in (d). If $|R_{\mathrm{anti}}(t_j)| < c^2 - 2v$ the paired eigenvalue e satisfies $|e| \le |R_{\mathrm{anti}}| + v < c^2 - v$, so $e \in G_v$. □

Remark 5.2. Part (d) explains the classical wrong-sign pathology quantitatively: the anti-balanced direction *sweeps the gap*, its free Rayleigh value traversing all of $(-c^2, c^2)$ as t runs over $(0, 1/\sqrt{2})$. The physical dual-balance sign of Ref. [6] is precisely the choice that avoids this sweep. The classical spurious states of equal-basis discretizations at $\kappa > 0$ [9, 10], exactly degenerate with the $-\kappa$ ground state [11], are a distinct, Coulomb-specific phenomenon outside the scope of Theorem 5.1; in the companion discretization they are reproduced at the 10^{-13} level [13].

6. The cusp lemma

The point-nucleus Coulomb–Dirac eigenfunctions carry the prefactor r^γ, $\gamma = \sqrt{\kappa^2 - (Z/c)^2}$. Consider the orthonormal generalized Laguerre family

with prefactor power μ,

$$q_j^{(\mu)}(x) = \sqrt{\frac{j!}{\Gamma(j+2\mu+1)}}\, x^{\mu} e^{-x/2} L_j^{(2\mu)}(x). \tag{11}$$

Theorem 6.1 (exact representation)**.** *For $\kappa < 0$ the lowest point-nucleus state of channel κ has both components proportional to the same profile $r^{\gamma}e^{-(Z/|\kappa|)r}$, with eigenvalue $E = \gamma c^2/|\kappa|$. Consequently, in the basis (11) with $\mu = \gamma$ and scale matched to $Z/|\kappa|$, the ground state lies in the span of $q_0^{(\gamma)}$ alone: the discretized problem reproduces it exactly with two degrees of freedom.*

Proof. Insert the ansatz $(a, b)\, r^{\gamma}e^{-\lambda r}$ into $h_\kappa\psi = E\psi$ with $V = -Z/r$. Each row produces one condition proportional to $r^{\gamma-1}$ and one to r^{γ}; the four conditions are linear in (a, b) and consistent precisely when $\lambda = Z/|\kappa|$ and $E = \gamma c^2/|\kappa|$, with a unique ratio b/a (for $\kappa = -1$, $b/a = Z/(c(\kappa - \gamma))$). The state is nodeless and matches the $n_r = 0$ entry of the Sommerfeld spectrum, hence is the ground state of the channel. Since $r^{\gamma}e^{-\lambda r} \propto q_0^{(\gamma)}(2\lambda r)$, both components lie in the one-dimensional span of $q_0^{(\gamma)}$. □

In the certified implementation this is observed literally: the two-degree-of-freedom discretization reproduces the ground binding energy of Z up to 130 at relative accuracy 10^{-15} [13].

Theorem 6.2 (universal rate dichotomy)**.** *Let $u(x) = x^{\gamma}e^{-x/2}$ with $\gamma > 0$, and let $c_j = \langle q_j^{(\mu)}, u\rangle$, $\mu > 0$. Then, exactly,*

$$c_j = \sqrt{\frac{j!}{\Gamma(j+2\mu+1)}}\, \frac{\Gamma(\mu+\gamma+1)\,\Gamma(j+\mu-\gamma)}{j!\,\Gamma(\mu-\gamma)}, \tag{12}$$

so that $|c_j| \sim \big[\Gamma(\mu+\gamma+1)/|\Gamma(\mu-\gamma)|\big]\, j^{-(1+\gamma)}$ as $j \to \infty$. In particular:

(a) *the decay exponent $-(1+\gamma)$ is independent of μ; a mismatched prefactor improves only the constant, never the rate;*
(b) *$c_j = 0$ for all $j \geq 1$ if and only if $\mu - \gamma \in \{0, -1, -2, \dots\}$; in particular the matched basis $\mu = \gamma$ represents u exactly;*
(c) *the L^2 tail after N terms scales as $N^{-(1/2+\gamma)}$.*

Proof. Formula (12) follows from the classical integral $\int_0^\infty e^{-x}x^{s}L_j^{(\alpha)}(x)\,dx = \Gamma(s+1)\Gamma(j+\alpha-s)/(j!\,\Gamma(\alpha-s))$ with $s = \mu+\gamma$, $\alpha = 2\mu$. The asymptotics follow from $\Gamma(j+a)/\Gamma(j+b) \sim j^{a-b}$: the normalization contributes $j^{-\mu}$ and the ratio $\Gamma(j+\mu-\gamma)/j!$ contributes $j^{\mu-\gamma-1}$, so the μ-dependence cancels in the exponent. The vanishing condition is the pole condition of $1/\Gamma(\mu-\gamma)$ together with positivity of the remaining factors, and (c) follows by summing $|c_j|^2 \sim j^{-2-2\gamma}$. □

Theorem 6.2 is the approximation-theoretic content of "exact structure or nothing": adaptive approximation of the cusp with any prefactor other

than γ itself faces the fixed algebraic rate $N^{-(1/2+\gamma)}$ in norm, no matter how the prefactor is tuned. This is consistent with, and explains, the operating accuracies of recent adaptive spectral solvers [12] as compared with the matched-basis solver of the companion paper [13]. The formula (12) is verified at quadrature precision (10^{-9}) across $\mu \in \{0.9, 1, 2\}$, the measured constant ratios agree with $\Gamma(\mu+\gamma+1)/|\Gamma(\mu-\gamma)|$ to five digits, and the measured decay slopes converge monotonically to $-(1+\gamma)$ [13].

7. Numerical evidence at the Coulomb frontier

Three measured facts delimit what remains to be proven. All were obtained with the certified discretization of [13] (run-time verification of the assembly identities; exact quadrature at the matched prefactor).

(i) Eigenvalue rate. For mismatched bases the ground-state eigenvalue error decays as $N^{-2\gamma}$: measured exponents $-1.901/-1.605/-0.928$ against $2\gamma = 1.913/1.624/0.966$ at $Z = 40/80/120$ ($\kappa = -1$, integer prefactor), with a *fractional* mismatched prefactor $\mu = \gamma + 0.3$ giving the same exponent (-1.593) at a different constant, exactly as Theorem 6.2(a) demands. The mechanism is quadratic in the form norm: the measured ratio $|\delta E_N|/\langle r_N, |H_0|\, r_N\rangle$, with r_N the projection residual of the exact eigenfunction, is 0.43–0.55 over the trustworthy window.

Conjecture 7.1 (first-order dual estimate)**.** *For the Galerkin approximation of an isolated eigenvalue of h_κ from a conforming family, $|\delta E_N| \le C\,\langle r_N, |H_0|\, r_N\rangle$ with C of order unity; combined with Theorem 6.2 this yields the $N^{-2\gamma}$ law.*

(ii) Relative bounds. The operator-norm relative bound $\|WH_0^{-1}\|$ and the form bound $\|\,|H_0|^{-1/2}W|H_0|^{-1/2}\|$ of the discretized Coulomb potential *diverge* with N ($\sim N^{0.46}$ and $\sim \log N$ respectively): no Kato–Hardy inequality can hold uniformly relative to the *discretized* free operator, because $(\Pi H_0 \Pi)^{-1} \neq \Pi H_0^{-1}\Pi$ — the discrete operator caps the available momentum at $\sigma_{\max} \approx 0.64\beta N$ while the basis keeps resolving the $1/r$ singularity. By contrast, the same form bound measured against the $|H_0|$ of a much larger nested reference — a proxy for the continuum operator — *converges* (e.g. $0.8633 \to 0.8687$ at $Z = 92$ with halving increments), remaining below 1 up to $Z/c \approx 0.75$ at practical basis sizes.

Conjecture 7.2 (continuum-relative protection)**.** *Formulated with the spectral projectors and the absolute value of the* continuum *free operator, the protection statement of Theorem 5.1(c) extends to the point-Coulomb potential for the balanced families in the matched basis, in the coupling range where the continuum-relative form bound is smaller than the fiber Rayleigh margin. The divergence of the discrete-relative bounds shows that no proof can proceed relative to the discretized free operator; compressions in the sense of Lewin–Séré [4] are necessary.*

(iii) The impostor's trajectory. At point Coulomb coupling the anti-balanced (sign-flipped dual-balance) discretization carries a spurious gap eigenvalue already at the smallest basis sizes and at every Z tested: it is born at the *lower* gap edge (binding -36905 at $Z = 1$, $N = 20$, i.e. $E \approx -0.965\,c^2$) and dives toward the gap midpoint as N grows ($-36905 \to -28314$ for $N = 20 \to 80$) — the sweep of R_{anti} of Theorem 5.1(d) made visible, accelerated by the unbounded potential. Matching this trajectory quantitatively is the natural sharpness test for Conjecture 7.2.

8. A conditional exclusion window at Coulomb coupling

The measured constants of Section 7 support a finite-dimensional exclusion theorem at point Coulomb coupling. Split the fibers at t^* into a low sector L ($t_j \le t^*$) and a high sector H ($t_j > t^*$), write $D^* = \sqrt{1 + 4t^{*2}}$, and let E_L, E_H denote the fiber energies on the full pair spans of each sector. Suppose the discretized Coulomb form satisfies the measured bounds $\|E_L^{-1/2} W_{LL} E_L^{-1/2}\| \le \omega_L$, $\|E_L^{-1/2} W_{LH} E_H^{-1/2}\| \le x$, and $\|E_H^{-1/2} W_{HH} E_H^{-1/2}\| \le \omega_H$.

Theorem 8.1 (conditional exclusion window)**.** *Let $\rho_H(m) = D^* c^2/(D^* c^2 - m)$ and suppose $\omega_H \rho_H(m) < 1$ and*

$$\Big[\omega_L + \frac{x^2 \rho_H(m)}{1 - \omega_H \rho_H(m)}\Big] \frac{c^2}{c^2 - m} < 1.$$

Then the full pair-space Galerkin matrix of $h = H_0 + W$ has no eigenvalue E with $|E| < m$.

Proof. On the high sector $|h_H| \ge D^* c^2$, so $\|E_H^{1/2}(h_H - \mu)^{-1} E_H^{1/2}\| \le \rho_H(m)$ for $|\mu| \le m$ (the function $E/(E - m)$ decreases in E on the spectrum); with the ω_H bound and a Neumann series, $h_H + W_{HH} - \mu$ is invertible with weighted resolvent bound $\rho_H/(1 - \omega_H \rho_H)$. The Schur complement onto L then carries a correction of weighted norm at most $x^2 \rho_H/(1 - \omega_H \rho_H)$, while $\|E_L^{1/2}(h_L - \mu)^{-1} E_L^{1/2}\| \le c^2/(c^2 - m)$ since $E_j \ge c^2$ on L. The displayed condition makes the Schur complement invertible, and the Feshbach block lemma yields invertibility of the full pencil at every $|\mu| < m$. □

At $Z = 40$ ($\nu = 0.29$, $t^* = 1$) the measured constants $(\omega_L, \omega_H, x) = (0.19, 0.35, 0.20)$ give $m^* \approx 0.70\,c^2$: the theorem excludes eigenvalues from the deep two-thirds of the upper gap, the physical ladder (which starts at $\gamma c^2 = 0.96\,c^2$) sits above the window, and the certified solver spectrum is verified to contain no eigenvalue inside it. This is, to our knowledge, the first quantitative pollution-exclusion window for a point-Coulomb Dirac discretization. At strong coupling ($Z \gtrsim 90$) the two-sector hypothesis fails through ω_H: the measured dyadic scale decomposition shows bounded within-scale blocks ($\Omega_{kk} \approx 0.35$–0.57, stable under basis growth) but flat cross-scale couplings (≈ 0.2 between *all* scale pairs — the scale invariance of $1/r$ in action), so the row sums grow logarithmically. Beating this flat

cross-scale structure with a multi-scale (Littlewood–Paley-type) elimination, using the analytic structure of the relativistic Hardy inequality rather than norm bounds alone, is the precisely delimited remaining step toward the unconditional theorem.

9. Concluding remarks

The fiber reduction turns kinetic-balance pollution from an observed phenomenon into computable algebra: exact leakage laws with exponents 3/1/3/1 and constants 1/1/1/2, an exact global defect, exact conformity of the untruncated dual-balance pair space, and a pollution dichotomy for bounded potentials in which the protected window and the locations of the guaranteed impostors are explicit. The cusp lemma supplies the complementary approximation statement with a universal exponent. What separates this elementary picture from the full point-Coulomb theorem is precisely quantified — and measured: the failure of discrete-relative bounds, the convergence of continuum-relative ones, and the impostor trajectory that any complete theory must reproduce. All numerical statements are reproducible from the open verification scripts accompanying the companion paper [13], and the core algebraic identities of Theorems 3.1, 4.1 and 5.1(a) — the balanced-leakage identity $t-\rho = 4t^3/(1+w)^2$, the per-fiber defect $g^2(1-g^2)$, and the in-gap characterization of the anti-balanced Rayleigh value with its mid-gap crossing at $t = 1/\sqrt{5}$ — have additionally been machine-verified in Lean 4 with Mathlib; the proof files accompany the repository.

References

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